README

BOB-164 — WCAG AA colour-contrast failures on the live dashboard

Revision: 2 **Last modified:** 2026-08-25T18:20:00Z

Round 2. Round 1's fix was correct at the token layer and introduced a regression at the *background* layer that neither oracle could see. This document records the fix for that regression, the oracle change that makes the whole class visible, and the corrections owed to round 1's claims. Every number below was produced by a command in this directory or by `npx ng test` in `frontend/`; none is carried over unmeasured.

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1. What round 1 got wrong, precisely

Round 1 raised `--color-text-secondary` for legibility as TEXT. That token is also painted as a BACKGROUND, and a change safe for one role is a regression in the other.

node	role of the token	pre-fix	after round 1
`.status.unknown` (`color: #ccc`)	fill	2.46:1	**1.09:1**
`.type-badge.unknown`	fill	1.93:1	1.13:1
`.quality-badge.unknown`	fill	1.93:1	1.13:1

Framing correction (§11.4.6). These were not passing nodes that the fix broke. All three were *already below the floor* before round 1 — `.status.unknown` at 2.46:1, both badges at 1.93:1. Round 1 made an existing, unseen defect substantially worse; it did not create it. The distinction matters because it locates the real failure in DISCOVERY, not in the round-1 edit.

Measurement correction. The round-2 review reported the badges as “white on that background, 3.95 → 1.74”. The rendered foreground is not white. Measured in a real headless Chromium against the shipped bundle (`measure_rendered_cascade.py`), `.type-badge.unknown` resolves to `rgb(169,183,198)` — `--color-text-primary`, inherited through `html, body` in `styles.scss` — giving 1.93:1, not 3.95:1. The class the review identified is real and its severity was understated.

2. Why both oracles were blind, and what changed

Neither blindness was an oversight; both were structural.

- **The arithmetic oracle** asserted text tokens against `SURFACES = [bgPrimary, bgSecondary, bgTertiary]`. A token used as a BACKGROUND was outside its pair set *by construction*.
- **The rendered-DOM oracle** scans a static dist with no backend, so the results table and the hooks list render EMPTY. The nodes simply were not on the page it measured.

Patching the four ratios would have fixed one instance. The class is removed instead: **`frontend/src/app/models/style-contrast.spec.ts`** reads the real stylesheets, extracts every declared `(foreground, background)` pair — resolving inherited foregrounds through the SCSS ancestor chain, same-file sibling rules, and the global `html, body` default — and asserts each across all sixteen palette × mode combinations. A token used as a fill is now inside the pair set by construction, and a fill added tomorrow is picked up without editing the spec.

Its honest boundary is stated in the file header and repeated here: it is a static extractor, not a CSS cascade engine; it holds every pair to 4.5:1 rather than inferring the 3:1 large-text exemption from CSS; and unresolvable values (`color-mix`, `rgba`, gradients) are SKIPPED, counted, and surfaced rather than silently passed.

3. The fix

The root cause is a ROLE error, not four bad numbers: a literal (`#fff`, `#ccc`, `#333`) cannot be correct against a fill that changes with the palette. Round 1 had already established the remedy for one fill with `onAccent`; round 2 generalises it.

new token	fill it sits on
`onAccentHover`	`accentHover`
`onSuccess` / `onInfo` / `onWarning` / `onDanger` / `onPurple`	the semantic fills
`onMuted`	`textSecondary`-as-a-fill
`onBorder`	`border`-as-a-fill

Every value is pure black or pure white — whichever contrasts more with that palette's fill. That always clears 4.5:1: `contrast(#000,F)` and `contrast(#fff,F)` are equal only where BOTH equal 4.58, so their maximum never drops below 4.58 and **no fill colour had to move**.

Also added: **`contrastText`**, an AA-pinned, hue-preserving text variant of `contrast`. The secondary brand gold was serving as the `code` foreground in the Jackett tables and measured 2.47:1 on nord/light — the same brand-as-text confusion round 1 fixed for `accent`, missed because no scan renders those tables.

Static badge fills (`.movie`, `.tv`, …) previously *inherited* `--color-text-primary`, which varies per palette; they now declare a palette-independent literal chosen against their own static fill.

4. RED → GREEN, measured like-for-like

Round 1 reported RED `74 failed / 479 passed (553)` against GREEN `652 passed (652)`. Those were produced by **different versions of the spec**, so the pair did not measure one change — the matching `479 passed` is what made it look continuous. Re-measured here with the FINAL oracle against the pre-fix sources (`git show HEAD:` for every non-spec file, specs kept current):

```
RED    Test Files   1 failed | 31 passed (32)         [+ 1 more
failing file]
       Tests      878 failed | 1736 passed (2614)
```

```
GREEN  Test Files  32 passed (32)
       Tests     2742 passed (2742)                    exit 0
```

The RED total breaks down as (counted from the per-test FAIL lines, not from assertion text, which vitest prints twice):

count	class
192	`accentText` / `dangerText` / `warningText` / `contrastText` absent at HEAD (4 tokens × 16 themes × 3 surfaces)
25	genuine sub-floor DECLARED TOKEN pairs — 23 `textSecondary`, 2 `textPrimary`
4	catalogue-level tests (runtime wiring, hue family, on-fill, no-hardcoded-foreground)
657	declared FOREGROUND-ON-FILL pairs (`style-contrast.spec.ts`, new this round)

The **25** reproduces the reviewer's independent count exactly. My total differs from the reviewer's 173 by 48 — precisely the `contrastText` assertions this round adds (16 themes × 3 surfaces).

The totals differ (2614 vs 2742) and that is expected, not hidden: the fix adds tokens, and new tokens create new test cases. A test count is not a like-for-like axis; the failure classes above are.

5. Rendered-DOM verification of the nodes the scan cannot reach

`measure_rendered_cascade.py` injects the exact markup `dashboard.component.html` emits, with the component's Angular style-scoping attribute, and reads `getComputedStyle` — a real cascade, not a static inference.

```
PRE-FIX  (shipped bundle)          9 of 11 measured nodes below
their floor
POST-FIX (fresh build, same theme) 0 of 11 measured nodes below
their floor
```

Full transcripts: `cascade_PREFIX_darcula_dark.txt`, `cascade_POSTFIX_darcula_dark.txt`.

6. The scanner was blind to axe's `incomplete` channel

`_summarise()` read only `violations`. axe reports `incomplete` when its own heuristics decline to decide — and a white-on-white node at 1.0:1, the most flagrant failure a contrast oracle can face, lands there with `messageKey: equalRatio`. Round 1 scored it as a PASS. That is a §11.4.201(6) false-null living inside the contrast oracle itself.

The channel is now read, adjudicated from the same WCAG formula, and counted in the exit code. axe abstaining does not mean the page is fine; where the colours resolve, this script decides.

`--self-check` proves the oracle can SEE before any clean run is believed (§11.4.107(10)):

```
  golden-bad  #golden-bad-equal    CAUGHT      (white on white,
via `incomplete`)
  golden-bad  #golden-bad-plain    CAUGHT      (ordinary sub-
floor, via `violations`)
  neg-control #negative-control    correctly silent
self-check: PASS
```

Glyph nodes, decided honestly. Post-fix the channel holds 40 nodes across 16 themes for three icon-glyph controls (`.bridge-retry`, `.theme-toggle`, `.caret`) where axe resolves NEITHER foreground NOR background, so no oracle here can compute a ratio. Reporting them as failures would be a §11.4.201(1) false-positive refusal; dropping them silently would rebuild the false-null. They are a NAMED class, `GLYPH_UNMEASURED`: printed every run, recorded in the JSON, excluded from the exit code, and **fenced** — a glyph node outside the checked-in allow-list still BLOCKS. Their SC 1.4.11 3:1 obligation remains genuinely unproven and is tracked as **BOB-184**, not claimed clean.

Whole-catalogue result on a fresh build:

```
TOTAL colour-contrast violation nodes across 16 theme(s): 0
TOTAL blocking `incomplete` nodes (FAIL / UNDECIDABLE /
GLYPH_FAIL): 0
TOTAL fenced GLYPH_UNMEASURED nodes (reported, not blocking): 40
```

7. Paired §1.1 mutations — every one caught

#	mutation	result
R2A	`.status.unknown` reverted to `color: #ccc`	CAUGHT — 13 failed
R2B	`onMuted` → a mid-grey that clears nothing	CAUGHT — 3 failed
R2C	extractor reads only `background-color`, dropping the shorthand	CAUGHT — control needle fired: 7 failed
R2D	`non-text` exclusion pointed at `.status`, a real text node	CAUGHT — the fence rejects a class the CSS does not support
R2E	a `code` site reverted to `var(--color-contrast)`	CAUGHT — 5 failed
R2F	scanner ignores the `incomplete` channel (round-1 behaviour)	CAUGHT — `#golden-bad-equal` MISSED → self-check FAIL
R2G	`.caret` removed from the glyph fence	CAUGHT — exit 1, 2 blocking nodes
R2H	footer token `--color-accent-text` → `--color-accent-typo`	CAUGHT *after* strengthening — see §8

R2C is the load-bearing one: a blinded extractor is reported as BLIND, not as clean. R2D closes the fence's real risk, which is not a stale entry but a false one.

Round 1's own mutation set was re-run against the CHANGED oracle to confirm it still bites (the round-1 review's `R1`–`R4` labels appear in no artifact in this repository, so the set re-run is the one round 1 documented as `M1`–`M4`; stated rather than guessed):

#	mutation	result
M1	`accentText` → `#9d001e` (the fix undone)	CAUGHT — 1 failed
M2		CAUGHT — hue guard, 1 failed

#	mutation	result
	`accentText` → `#c4c4c4` (a grey that clears the ratio)	
M3	`onAccent` → `#000000`	CAUGHT — 1 failed
M4	`textSecondary` → `#808080`	CAUGHT — **3** failed

M4 now trips three assertions rather than one: reverting the token is caught by the surface-pair oracle as before, and additionally by the new fill-pair oracle, which is exactly the coverage that was missing.

Exit statuses were read directly from `$?` with no pipe in between (§11.4.201(12)).

8. Corrections owed to round 1’s claims

- **“NEW failures introduced by the fix: 0”** (round 1 §5) — **false as stated**. The fix drove `.status.unknown` from 2.46:1 to 1.09:1. The claim was true only *within the 30 nodes axe rendered*, and it was written without that scope. Scope now stated everywhere it appears.
- **“Previously-passing nodes now reported failing: 0”** (round 1 §7) — true only of the same 30-node scanned view, which excluded every data-driven node. Re-stated with its scope.
- **“TWO INDEPENDENT ORACLES (§11.4.245 structural independence)” — overstated for the DOM path.** `axe_contrast_scan.py` takes `fgColor` / `bgColor` / `shadowColor` FROM axe and re-does only the arithmetic; that is *arithmetic* independence, not *measurement* independence, and a wrong colour read by axe would be reproduced faithfully by the Python side. The vitest spec IS genuinely independent for declared pairs — it resolves colours from the palette source, never from a browser. The file header now says exactly this.
- **The `accentHover` sub-floor finding** (round 1, recorded unfixed in four palettes) is **fixed**, not filed: `onAccentHover` covers that fill and the generalised on-fill test asserts it in all 16 combinations. The round-1 note claiming otherwise is removed with it.
- **The footer regex** `var\(--color-accent[a-z-]*\)` accepted tokens that do not exist. Narrowing it to a closed alternation was *not enough* — the paired mutation still passed, because a valid sibling token satisfied an “at least one” check. The gate now asserts that

every `--color-*` the footer references is a real key in `TOKEN_CSS_VAR`; R2H then fails as it should.

9. What this evidence does NOT cover

- **Legibility is not contrast.** No human has looked at these colours. §11.4.185 manual QA is unaffected and still owed.
- **The served bundle is still stale.** `download-proxy/src/ui/dist/` carries the pre-fix defect verbatim; `scripts/install.sh:133` checks only that the directory EXISTS, never that it is newer than its sources, and `dist/` is gitignored so the divergence never shows in a diff. §11.4.108 layer 2 is **not** closed. Tracked as **BOB-183**; deliberately not rebuilt here (a sibling stream owns that tree).
- **Data-driven nodes are arithmetic-only in a rendered DOM.** The badge and status nodes are asserted by `style-contrast.spec.ts` and were measured once by `measure_rendered_cascade.py`, but the standing scan still cannot see them. Tracked as **BOB-185**.
- **Icon glyphs are unmeasured**, per §6. Tracked as **BOB-184**.
- **Resting state only.** Hover states are now covered where a stylesheet declares them (`onAccentHover`), but focus, active and error states were not rendered.
- **`/jackett/*` was not scanned in a rendered DOM**; its tokens are covered by the arithmetic oracles only.

10. Reproducing

```
# both arithmetic oracles
cd frontend && npx ng test --watch=false          # 2742 passed,
        exit 0

# the rendered-DOM oracle, validated before use
.venv/bin/python docs/qa/BOB-164/axe_contrast_scan.py --self-check
cd frontend && npx ng build --output-path=/tmp/dist_r2 && cd ..
.venv/bin/python docs/qa/BOB-164/axe_contrast_scan.py \
    --dist /tmp/dist_r2/browser --all-palettes \
    --out docs/qa/BOB-164/scan_GREEN_round2_all_palettes.json   #
        exit 0

# the rendered cascade for nodes the scan cannot reach
.venv/bin/python docs/qa/BOB-164/measure_rendered_cascade.py --
        dist /tmp/dist_r2/browser
```

Files

File	What
`axe_contrast_scan.py`	rendered-DOM oracle; reads `violations` AND `incomplete`; `--self-check` validates it against golden-good/golden-bad fixtures
`measure_rendered_cascade.py`	measures the real cascade for badge/status nodes the scan cannot render
`cascade_PREFIX_darcula_dark.txt` / `cascade_POSTFIX_darcula_dark.txt`	those measurements, 9/11 failing → 0/11
`scan_GREEN_round2_all_palettes.json`	post-fix, all 16 themes — 0 violations, 0 blocking incomplete
`scan_RED_shipped.json` / `scan_RED_all_palettes.json`	round-1 pre-fix scans, retained
`scan_GREEN_darcula.json` / `scan_GREEN_all_palettes.json`	round-1 post-fix scans, retained
`frontend/src/app/models/palette.contrast.spec.ts`	declared TOKEN-PAIR oracle (text on surfaces, on-fill tokens, hue guard)
`frontend/src/app/models/style-contrast.spec.ts`	declared FILL-PAIR oracle — closes the background-role blindness